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Designing a practical fatigue detection system: A review on recent developments and challenges

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ABSTRACT

Introduction: Fatigue is considered to have a life-threatening effect on human health and it has been an active field of research in different sectors. Deploying wearable physiological sensors helps to detect the level of fatigue objectively without any concern of bias in subjective assessment and interfering with work.
Methods: This paper provides an in-depth review of fatigue detection approaches using physiological signals to pinpoint their main achievements, identify research gaps, and recommend avenues for future research. The review results are presented under three headings, including: signal modality, experimental environments, and fatigue detection models. Fatigue detection studies are first divided based on signal modality into uni-modal and multi-modal approaches. Then, the experimental environments utilized for fatigue data collection are critically analyzed. At the end, the machine learning models used for the classification of fatigue state are reviewed.
Practical Applications: The directions for future research are provided based on critical analysis of past studies. Finally, the challenges of objective fatigue detection in the real-world scenario are discussed.

1. Introduction

Fatigue resulted from excessive physical or mental workload is one of the most significant occupational health and safety risks in many hazardous industries (Morrow & Crum, 2004; Fang et al., 2015; Jepsen et al., 2015). Fatigue can result in serious short- and long-term health/safety issues (Maman et al., 2017; Lock et al., 2018; SafeWork, 2013; WorkSafe, 2020). In the short term, fatigue results in impaired attention and reduced alertness as the main cause of injuries and accidents (Lerman et al., 2012; Williamson et al., 2011). It is estimated that between 70–90% of accidents and incidents are due to human errors (Safe Work Australia, 2015; NOPSEMA, 2020), for which fatigue is the main contributor (Theophilus et al., 2017; Love et al., 2019). In the long term, fatigue results in severe health conditions such as heart disease, diabetes, depression, and burnout as the byproduct of long-term or chronic fatigue (WorkSafe, 2020; Van der Hulst, 2003).

Due to the significant impact of fatigue on safety and well-being, there are several fatigue studies conducted in various domains such as psychology, transportation, construction, and medicine (Sikander & Anwar, 2019; Van Cutsem et al., 2017; Lee et al., 2016; Chang et al., 2009). Although fatigue has been an active field of study for quite a long time, fatigue assessment has mainly been done using subjective assessment methods. If we look at how fatigue is defined in previous studies, most of them defined fatigue as the subjective feeling of exhaustion, tiredness, or discomfort (Piper et al., 1987; Hardy et al., 1997; Doris et al., 2010; Hirshkowitz, 2013). Over the years, many well-known subjective assessment approaches like NASA-TLX, Self-report questionnaires, Chalder fatigue scale, Likert scale, Karolinska Sleepiness Scale (KSS), Borg's RPE, and Stanford Sleepiness Scale have been used to assess fatigue across different domains (Dittner et al., 2004; Shen et al., 2006; Chalder et al., 1993). Although these subjective methods are popular and cost-effective to assess fatigue, they are not often fully reliable due to the fact that the subjective assessment can be biased, which can lead to incorrect fatigue assessment. To overcome this problem, recent fatigue assessment studies have used physiological signals such as electroencephalogram (EEG), electrocardiogram (ECG), electromyography (EMG), heart rate (HR), and motion to objectively assess fatigue (Martins et al., 2021; Van Cutsem et al., 2017; Sikander & Anwar, 2019; Charles & Nixon, 2019; Mascord & Heath, 1992). The monitoring of physiological signals offers valuable insights into the electrical activity of a particular body region. Identifying any anomalous variations in these data can aid in detecting fatigue (Faust et al., 2018).

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The emergence of wearable technology has greatly facilitated the acquisition of physiological data, making objective fatigue detection more feasible than ever before. As a result, the use of such technology has gained considerable attention in recent research on fatigue assessment.

There are a number of past review studies focused on the emerging field of objective fatigue measurement using physiological sensing. Most of these studies have been conducted in the domain of driver fatigue monitoring (Sikander & Anwar, 2019; Abbas & Alsheddy, 2021; Ibrahim et al., 2022; Kamti & Iqbal, 2022). In addition, there have been domain-specific reviews in fields such as occupational industry, medicine, and sports(Sun et al., 2022; Marotta et al., 2022; Block et al., 2022). Although past review papers have focused on areas such as strengths and limitations of different physiological parameters, signal processing, significant feature matrices, and so forth, several key aspects such as signal modality, experimental environment and machine learning model were not extensively analyzed. In this review, we aim to conduct an in-depth review of objective fatigue detection studies with respect to: (a) signal modality, (b) experimental environment, and (c) machine learning models used for the classification of fatigue state. Signal modality is a significant topic that will provide the trends of current practices and benefits or disadvantages of choosing a modality. Experimental environment is another important part for any study for assessing the deployability, including user experience, safety, and non-intrusiveness of the proposed system. Finally, selection of machine learning model to predict fatigue state is a vital step in achieving best the accuracy and efficiency based on recent reported successes. Considering these important aspects, this review will provide valuable insight in designing practical fatigue detection and monitoring system.

In this study, we aim to analyze state of the art studies in fatigue detection using physiological signals to find the current trend and shortcomings. We have a primary research question followed by a set of sub-questions that will be used to collect findings from the selected articles for further discussion. The research questions are the following:

Primary research question: What are the current fatigue detection approaches which utilize physiological signals?

Q1. What modality of physiological signal is used in fatigue detection?
Q2. What experimental environments are utilized for fatigue data collection?
Q3. Which machine learning models are used for the classification of fatigue state?

This review aims to critically analyze the state-of-the-art fatigue detection approaches proposed within the past five years. The primary objective of this study is to provide a comprehensive assessment of the technologies utilized for detecting fatigue. This will aid in obtaining a thorough understanding of the current research landscape and identifying potential gaps that require further investigation. The focus of this study will be on (1) evaluating the studies based on the type of employed signal modality to determine if there exists a bias in signal selection; (2) investigating various experimental environments for data collection to determine if factors such as experiment duration may have an impact on the obtained fatigue data; and (3) providing an in-depth analysis of different fatigue classification models. By the end of this review, all limitations present in existing studies, ranging from data collection methodologies to fatigue detection approaches, will be mapped and discussed. Finally, recommendations for future research directions will be provided based on the identified gaps and limitations.

2. Research method

The following subsections discuss the methods utilized in this review. In this article, we utilized a systematic approach in collecting past studies for further analysis.

2.1. Review process

Figure 1 presents the complete flow of the review process from start to end. The process started by defining the research topic, which is the target focus of the study. The next process was to define the research question on the target topic, which allows setting a clear focus by narrowing the topic. One of the key processes after defining the research question is to devise the search strategy as this allows to find the best set of articles that potentially will be selected for the review. The following section elaborates on the search techniques used in this study.

2.1.1. Search strategy

A systematic search approach was used to find relevant studies based on the research questions. Three sets of keywords were derived from the research questions that subsequently was used for searching. A complete list of keywords based on the topic is presented in **Table 1**. The key search term is set to *fatigue* to identify all the articles related to fatigue. As the key aspect of this review is regarding the fatigue detection approach, all the related terms of *detection* are used to expand the searching. The last set of keywords is derived based on *physiological signals*, which is the secondary aspect of the topic to narrow down the search results.

Three electronic databases (Scopus, Web of Science (WOS) and IEEE Explore) were used for article searching using the defined keywords. A detailed search strategy based on databases is shown in **Table 2**.

2.1.2. Inclusion/exclusion criteria

From the selected databases, a total of 3,371 articles were retrieved using the keywords presented in **Table 2**. Additional refinement was added to each database by limiting the publication date to the last five years (2018–2022), English as language, and limit it to journal articles. This resulted in the retrieval of 729 articles for the next screening phase. To identify the relevant articles for the target topic, a set of inclusion and exclusion rules was defined as listed in **Fig. 1**. After reading the title and abstract using the inclusion/exclusion rules, 294 articles were excluded, leaving 435 articles to be selected using the full text. As many articles are indexed by multiple databases, it is highly possible to have many duplications among the collected articles. A total of 197 duplicate articles from the collection were found, and, after removal, we obtained 238 studies for further screening.

2.1.3. Quality assessment

In order to select high quality studies, additional selection criteria were implemented that were adopted from Martins et al. (2021) as listed in **Table 3**. Although a set of seven criteria was listed, we decided to exclude the fifth. In their fifth criteria, the authors mentioned the validity to use the wearable device for the signal collection that has been used in at least one peer-reviewed journal published in the past. However, identifying each device from 238 studies and subsequently searching for previously published studies that used them is not feasible for this review. Therefore, we decided to exclude this criterion from the selection criteria. After reading the full-text papers based on the selection criteria, another criterion (balanced dataset) was found not to be ideal for the screening since there are only few studies providing the required information regarding this criterion. Therefore, this criterion

Table 1
List of search keywords used in this study for each topic.

Topic	Keywords
Fatigue	fatigue
detection	monitor*, detect*, measure*, assess*, screen*, identi*, predict*, estimat*
physiological signals	”wearable sensor*”, ”sensor*”, ”physiological signal*”, signal, EEG, Electroencephalogram, ECG, Electrocardiogram, EMG, Electromyography, motion, ”heart rate”, HRV, HR, temperature, EOG, Electrooculogram, gait

Table 2
List of databases, search queries and the number of articles (Hit), which was found by executing the specific query in the specific database.

Database	Search Query	Hit
Scopus	(TITLE(fatigue AND (monitor* OR detect* OR measure* OR assess* OR screen* OR identi* OR predict* OR estimat*) AND ("wearable sensor*" OR "sensor*" OR "physiological signal*"OR signal OR EEG OR Electroencephalogram OR ECG OR Electrocardiogram OR EMG OR Electromyography OR motion OR "heart rate" OR HRV OR HR OR temperature OR EOG OR Electrooculogram OR gait)))	1441
WOS	(fatigue AND (monitor* OR detect* OR measure* OR assess* OR screen* OR identi* OR predict* OR estimat*) AND ("wearable sensor*" OR "sensor*" OR "physiological signal*"OR signal OR EEG OR Electroencephalogram OR ECG OR Electrocardiogram OR EMG OR Electromyography OR motion OR "heart rate" OR HRV OR HR OR temperature OR EOG OR Electrooculogram OR gait))	1118
IEEE Xplore	("Document Title":fatigue) AND ("Document Title":monitor* OR detect* OR measure* OR assess* OR screen* OR identi* OR predict* OR estimat*) AND ("Document Title":"wearable sensor*" OR "Document Title":"sensor*" OR "Document Title":"physiological signal*"OR signal OR "Document Title": EEG OR "Document Title":Electroencephalogram OR "Document Title":ECG OR "Document Title": Electrocardiogram OR "Document Title":EMG OR "Document Title":Electromyography OR "Document Title":motion OR "Document Title":"heart rate" OR "Document Title":HRV OR "Document Title":HR OR "Document Title":temperature OR "Document Title":EOG OR "Document Title": Electrooculogram OR "Document Title":gait)	812

was also removed. Among 238 studies, an additional 48 papers were found not relevant to the topic according to inclusion/exclusion criteria and were removed. The remaining five criteria (**Table 3**) were used to further screen the remained 190 papers to identify high-quality studies.

While gathering meta-data from each study using aforementioned five criteria, it was found that not all studies can satisfy the 'cross-validaiton' criteria. Cross-validation methods to get an unbiased estimation of the fatigue detection model were found to only be applicable to the studies that used machine learning-based models. Among the 190 studies, 100 articles used the statistical approach to identify and analyze fatigue, while the rest of the studies used machine learning models for fatigue detection. Due to this, the articles were further grouped into two groups, including machine learning and statistical methods. We excluded statistical studies since they cannot satisfy the cross-validation criteria. Machine learning-based studies were screened based on the above-mentioned five criteria, and only studies that scored five out of five were included in the final selected studies. A detailed ranking of the studies is listed in **Table 4**.

Finally, 43 articles highlighted in Table 4 were selected as the final set of articles for further review and analysis. By implementing the specified selection criteria, the articles that are accepted demonstrate a higher level of quality, as they score above the predetermined threshold for quality assessment. This approach ensures that the articles selected adhere to rigorous standards.

2.2. Selection result

Fig. 2 illustrates the trend of research conducted over the last five years, showing a significant increase in using machine learning to detect fatigue. Although statistical methods were the most established and used means by the fatigue research community, machine learning approaches are rapidly taking over. The research community is embracing the use of machine learning techniques due to its greater generalization ability in detection of fatigue, ability to handle large dataset, automate the fatigue monitoring system by predicting fatigue state, and many more.

Fig. 3a shows the target population of the selected fatigue detection studies. Most of the research focused on the detection of fatigue in drivers followed by workers and other occupations. In addition, it is observed that EEG is the most popular physiological signal used to detect fatigue (**Fig. 3b**). Although EEG is used in a significant number of studies, it was primarily aimed at detecting mental fatigue rather than physical fatigue. ECG, EMG, and motion/gait signals were used in different studies for physical fatigue detection.

Over the last five years articles have been published in several disciplines like engineering, computer science, physics and astronomy, medicine, biochemistry, genetics and molecular biology, and many more. **Fig. 4** illustrates the leading journals which published more than one articles in fatigue detection. Among these, *IEEE ACCESSS, and SENSORS* were the top sources publishing six and four articles, respectively, over the period last five years.

In the following section, we have reviewed the selected 43 studies. A summary table of the selected studies is listed in **Table 5**.

3. Signal modality

Fatigue detection using single physiological signal is found to be the most popular approach among the selected studies. Only 8 out of 43 studies (Table 5) implemented more than one physiological signal in their study.

3.1. Uni modal

Based on the type of fatigue (mental/physical) we grouped unimodal studies into two parts. In the first part, we discuss the studies that focused on mental fatigue and in the subsequent part, we focus on the studies targetimg physical fatigue (Table 5). Using several selection criteria as discussed in the previous section, it can be observed that the number of final selected studies that targeted mental fatigue are significantly higher than physical fatigue studies.

Among all the uni modal approaches that focused on mental fatigue, EEG is the found to be dominating (27 out of 35). Table 6 lists the studies using EEG to detect mental fatigue. Unlike other physiological signals such as ECG, gait, and motion, EEG is only suitable to detect mental fatigue. To collect the mental fatigue and attention related data using EEG device, most of the studies utilized multiple channels in the areas like fronto-polar (Fp), frontal (F), central (C), temporal (T), parietal (P), and occipital (O) of the cerebral cortex following the International 10–20 system for the placement of electrodes. Although in some studies, the dry electrode (Wang et al., 2021; Liu et al., 2021; Wang et al., 2021; Xu et al., 2021) and wireless EEG devices (Wang et al., 2021; Ko et al., 2020; Min et al., 2022; Zhang et al., 2021a; Xu et al., 2021) were utilized, most of the studies used the traditional wet electrodes EEG device to collect the brain signals. Increase in the number of channels will increase both cost and computational complexity of processing the signal and building fatigue detection system. To overcome this issue, few studies tried to investigate the fatigue detection performance using a less number of channels or certain regions of brain. According to the several studies (Wang et al., 2021; Xu et al., 2021; Wang et al., 2021; Liu et al., 2021; Luo et al., 2019), the most prominent areas for attention and cognitive vigilance are the frontal and occipital area. Sheykhivand et al. (2022) investigated six active regions among which one region contained only one electrode (T6) that showed promising accuracy when compared with several state of the art network based models. Wang et al. (2021) found the detection performance of 10 channel's Common Spatial Pattern (CSP) features in frontal region achieved a close accuracy in-comparison to all 14 EEG channel's features. Liu et al. (2021) investigated 24 channels of EEG data in pairs but found three channels in non–hair bearing region achieved the highest accuracy of 90.12%. Interestingly, all these studies investigating the use of fewer number of channels are targeting drivers' mental fatigue detection. By analyzing the past studies, no significant correlation was found between the performance of fatigue detection and number of channels. The possible reason for this is the use of various features extraction approaches used

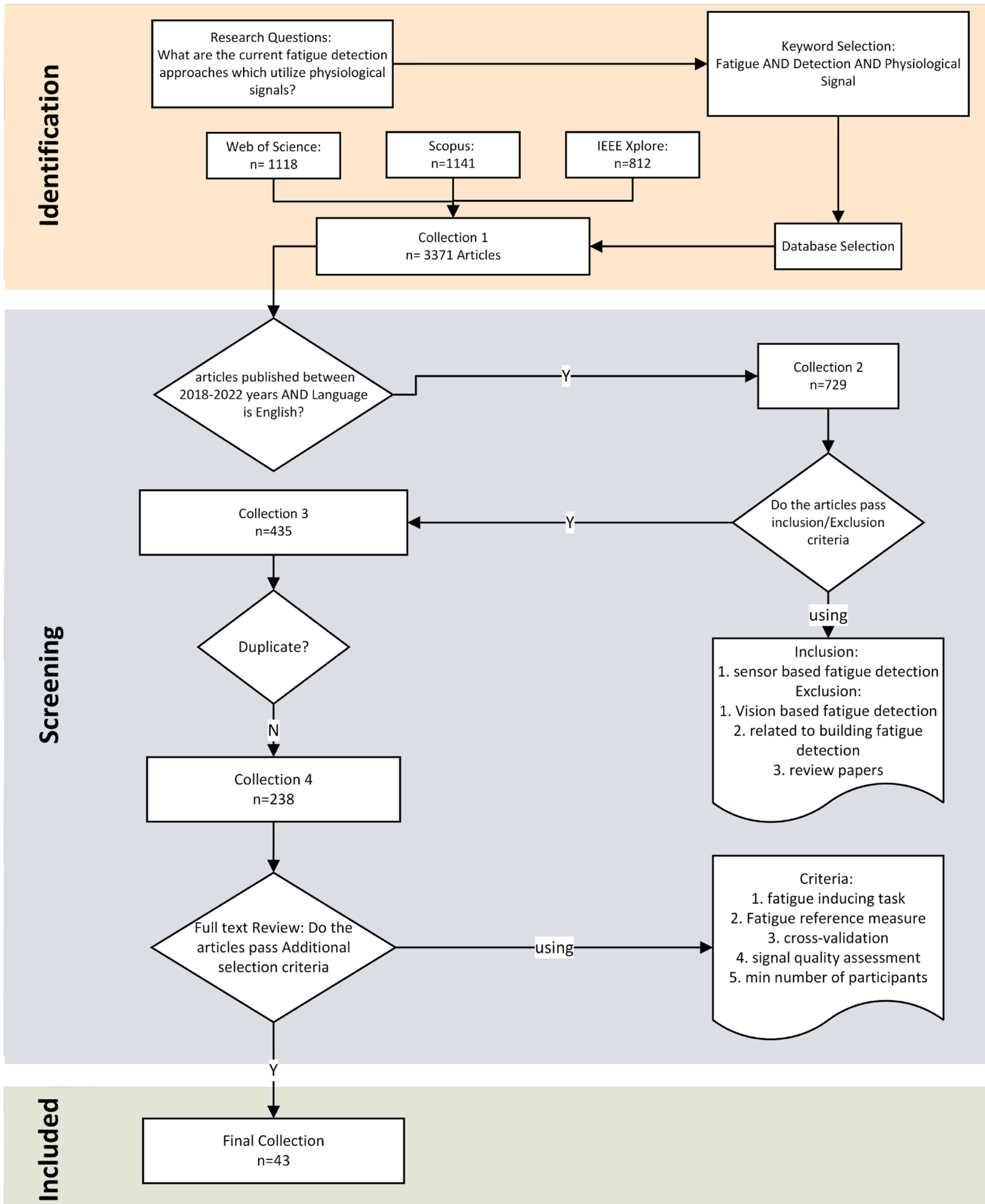

Fig. 1. Review process flow chart.

by the studies. The most commonly used feature extraction methods are power spectrum dynamics (PSD), common spatial pattern(CSP), convolutional neural network (CNN), and entropies. Studies using neural network does not require traditional manual feature extraction methods as they can learn and identify important features from the raw dataset. Thus it is difficult to discuss which features neural network finds important due to lack of interpretability. On the other hand, studies that used manual feature extraction are dependent on the set of features to be fed to the model, which can also be questionable as they can be biased to achieved better recognition by the model. Therefore, it is not possible to conclude which features are significant due to the variability of the classifier as well as the features.

Apart from EEG, ECG (Pan et al., 2021) and EOG (Kolodziej et al., 2020) were also considered in the identification of the mental fatigue state. Pan et al. (2021) used principal component analysis to extract features from the ECG signal to analyze the mental fatigue state of the pilots that showed a good result. The authors stated that sympathetic and parasympathetic nerves composing the autonomous nervous system (ANS) have significant changes due to mental fatigue. Thus the heart rhythm which is a part of ANS does contain significant mental fatigue information for investigation.

Although a majority of the studies focused on the mental fatigue detection, the research on physical fatigue detection was also found to be prominent. Several studies investigated physical fatigue using EMG, ECG, gait, and kinetic data of the participants in an experimental setup using exercises such as pilates, running, walking, or manual handling of objects. In contrast to the majority of the studies on driver's mental fatigue or drowsiness detection using EEG, Naim et al. (2022) used EMG to detect muscle fatigue of the drivers. The study utilized a combined time-domain and frequency-domain features extracted from EMG signal to detect muscle fatigue that aimed to be non-intrusive, unlike EEG based fatigue monitoring system. Ni et al. (2022) extracted 24 HRV

Table 3
Quality assessment - Additional selection criteria to select studies.

Criteria	Risk factor	Martins et al. (2021)	this study
Fatigue inducement	Fatigue was not induced	✓	✓
Validity of Reference	Biased reference measure of fatigue	✓	✓
Balanced dataset	# samples in non-fatigued state # samples in fatigued state	✓	
Cross-validation	Overfitting or overestimation of model performance	✓	✓
Validity of outcomes data	Unreliable input data	✓	
Signal Quality Assessment	Unreliable input data	✓	✓
Number of participants	Limited generalizability	✓	✓

Table 4
The rating of 'machine learning' and 'statistical model' studies based on quality assessment criterion listed on Table 3.

Rating	Number of machine learning studies	Number of statistical model studies
5*	**43**	0
4	32	33
3	12	49
2	3	16
1	0	2
Total	90	100

*5 shows the highest quality papers selected for the current review paper.

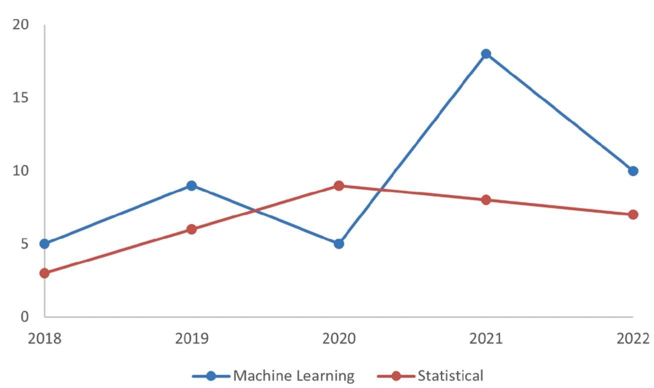

Fig. 2. Trend of published number of articles presenting machine learning and statistical models in fatigue detection in last five years.

features from ECG signal to classify physical fatigue to reduce exercise related fatigue. Kadry et al. (2022) and Jiang et al. (2021) both used inertial measurement unit (IMU) sensors to collect acceleration and angular velocity of the subjects for the analysis of gait to detect physical fatigue. Using IMU placed on the ankle, Baghdadi et al. (2018) collected the motion data such as jerk, acceleration, velocity, position, and posture to analyze physical fatigue. Apart from IMU, Aoki et al. (2021) used a kinetic sensor to collect gait behavior in both non-fatigue and fatigued conditions. Unlike mental fatigue studies, studies targeting physical fatigue detection has more potential in providing practical and non-intrusive solution due to the availability of various lightweight devices.

3.2. Multi modal

Multi-modal fatigue detection is reported to have the advantage over uni-modal approaches to be robust and achieve better accuracy (Craye et al., 2016; Wang et al., 2014). Eight studies out of 43 studies used more than one physiological signal to detect fatigue (Table 5). Unlike uni-modal approaches, the use of multi-modal approach to fatigue detection is yet in the early stages. According to the literature, multi-modal approach is a better solution in identifying the manifestation of fatigue due to the increasing volume of physiological data. Due to the lack of concusses in the manifestation of fatigue, the use of more than one physiological parameter can enhance the possibility of capturing valuable data related to fatigue that may have been excluded in uni-modal data. However, using multiple devices to capture the signals can be a hurdle to propose practical and non-intrusive fatigue monitoring solutions. In a working environment where high mobility is needed, using multiple devices is not only a matter of acceptance by the participant but also is subject to data synchronization related challenges. Collecting and processing signals from multi-source is complicated, which subsequently is a hurdle for the practical fatigue monitoring system.

To study the mental fatigue state, several studies used EEG in combination with other physiological signals. Min et al. (2022) used EEG and video image of eyeblink. Wang et al. (2019) and Zhang et al. (2022) used EEG and EOG signals to detect the mental fatigue state of the drivers. Ramírez-Moreno et al. (2021) used EEG, HR, HRV, EDA and ST signals for workplace mental fatigue monitoring. To identify ship navigator's mental fatigue, Monteiro et al. (2020) used EEG, ECG, ECG, EMG, ST, and eye-tracking signals. To find driver's different mental conditions such as fatigue, stress, and drowsiness, Choi et al. (2018) was the only study that did not use an EEG signal. Rather they used PPG, GSR, temperature (TEM), acceleration (ACCEL), and the rate of rotation (GYRO) signals for the multi-class classification of the driver's mental condition. The use of EEG in fusion with other physiological signals to identify effective mental fatigue solutions was found to be the priority

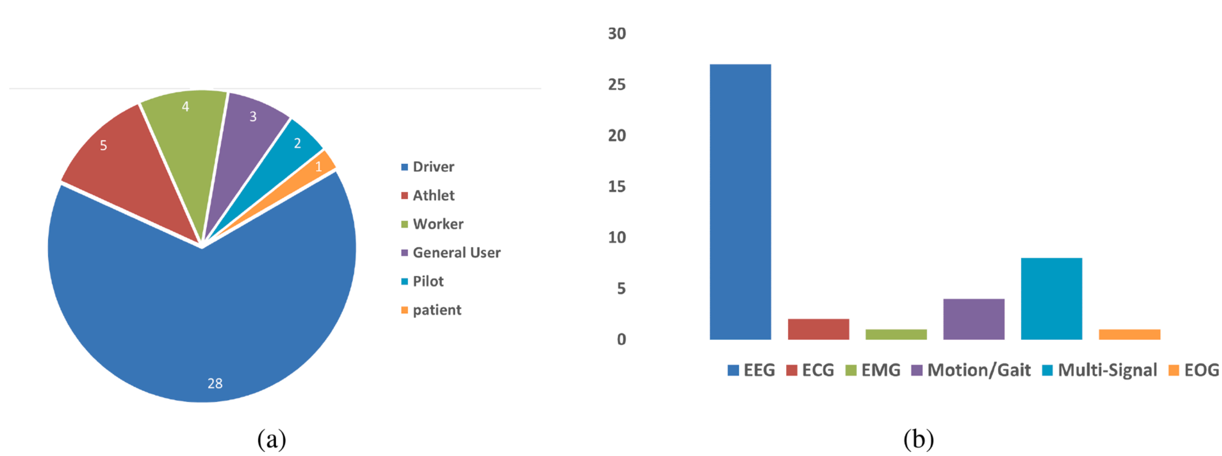

Fig. 3. Distribution of selected articles based on: (a) target population; and (b) physiological signal/s used for fatigue detection.

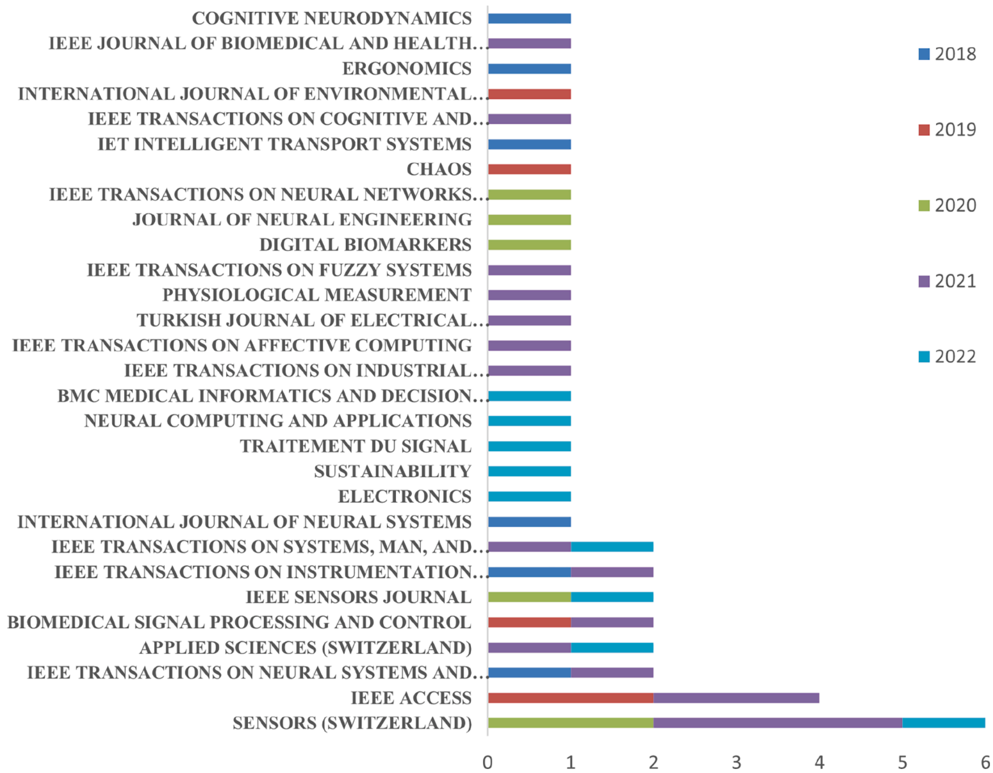

Fig. 4. List of Journals that published fatigue detection articles in last five years.

even among the multi-modal studies. This also raises the question of whether the EEG features were more significant in achieving fatigue detection accuracy than other signal features.

To study exercise-induced physical fatigue, Li and Chen (2022) collected EMG and ECG signals. In contrast to all the multi-modal studies, Luo et al. (2020) collected accelerometer, PPG, ST, galvanic skin response, barometer signals from the participants to study both physical and mental fatigue state.

4. Experimental environment

After reviewing the past studies, we identified three types of research environments including simulated environment, controlled environment, and real environment for physiological signal collection. The studies under simulated and controlled environments basically utilized lab facilities to conduct the experimental tests. The experimental protocol of those studies, which used the simulation software for visualization and hardware for intervention to mimic certain tasks, were grouped under simulated environment. On the other hand, studies targeting manual tasks such as exercise and material handling experimental protocol in lab were grouped under controlled environment. Lastly, studies that conducted the experimental protocol out of the lab environment and considered real world uncontrolled environment were grouped under real environment. The following section discusses various data collection protocols in terms of type and duration of experimental task under each group.

4.1. Simulated environment

There are 34 studies out of 43 conducted in a simulated environment. The simulated environment studies are listed in Table 7 and are analyzed based on type and duration of the experiment. As shown, the past studies have been grouped into driving, flight, ship, auditory/visual stimuli and PVT. A total of 27 studies focusing on driver fatigue detection adapted a driving simulation experiment to simulate the real driving situation in lab environment. Among these, 24 studies collected EEG data from the simulated driving scenario. Studies targeting pilot and ship navigator mental health detection also used simulated environments such as flight simulation and ship navigation. As shown in Table 7, to induce fatigue from driving simulation, flight simulation, and ship navigation protocol, different studies have taken different lengths of sessions ranging from 30 to 360 min. In driving simulation the majority of the articles conducted the experiment in the range between 40–360 min, indicating the slow induction of fatigue in such setup. However, Wang et al. (2021) conducted the driving sessions only for 30 min and argued that this period is sufficient to induce fatigue in driving simulation. Similar to driving simulation, studies that used flight simulation protocol conducted the experiment for longer period (120–360 min). The reason for such longer experiment is to investigate the impact of long flight time as the cause of pilot fatigue. Apart from all, Luo et al. (2019) did not follow any fixed time, and instead they relied on the verbal report of fatigue by the participant to end the experiment session.

Most of the studies that took a simulation setup such as driving, flight, and ship navigation to reach a recognizable level of fatigue, recorded a self-assessed level of fatigue by the participants for data segmentation based on alert and fatigue state. Typically in these experimental setups, the initial experimental period of 5 min to 30 min is considered as alert or normal state, whereas the last 5 to 10 min of the experiment are considered a fatigue state. Although making such assumption makes the data processing less complicated, it will raise the question of bias in data labeling. The reason is that the duration of fatigue manifestation of different subjects are different due to their physiological difference as well as other personal factors such as sleep-quality, life-style, and environmental factors. Due to these challenges, several studies relied on the participants self-assessed level of fatigue to segment data.

Apart from driver fatigue, pilot fatigue, and ship navigator fatigue studies, several studies used auditory stimuli (Ramírez-Moreno et al., 2021), visual stimuli (Peng et al., 2021), Psychomotor vigilance test

Table 5
Summary of selected (43) fatigue detection studies.

Signal	Fatigue inducing task	Fatigue reference measure	Participants	Cross-validation	Target	Reference
EEGEEG	Driving SimulationDriving Simulation	9-point KSS	16	10-fold cross-validation	DriverDriver	Dang et al. (2021)
		9-point KSS	10	Tenfold cross-validation		Gao et al. (2019)
		9-point KSS	10	Leave-one-out cross-validation		Cai et al. (2019)
		9-point KSS	11	50% training, 50% testing		Yang et al. (2021)
		Borg's CR-10	16	80% training set and 20% of the test set		Luo et al. (2019)
		Chalder fatigue and Lee fatigue scales	11	50% training, 10% validation, 40% testing		Sheykhivand et al. (2022)
		Chalder Fatigue and Lee Fatigue questionnaires	11	5-fold cross-validation		Sheykhivand et al. (2022)
		EOG, self-reported fatigue,	22	10-fold cross validation		Hu and Min (2018)
		KSS	8	10-fold cross validation		Gao et al. (2019)
		Li's subjective fatigue scale and the Chalder fatigue scale	12	Ten-fold cross-validation		Wang et al. (2018)
		NASA-TLX	13	cross-subject cross-validation		Zeng et al. (2021)
		NASA-TLX	34	The grid search CV (5-fold cross validation)		Wang et al. (2021)
		NASA-TLX	15	15-fold cross-validation		Zeng et al. (2020)
		RT	27	leave-one-subject-out strategy		Zhang et al. (2021b)
		RT	31	10-fold cross validation		Xu et al. (2021)
		RT	20	five fold cross-validation		Wang et al. (2021)
		RT	29	five fold cross-validation		Wang et al. (2021)
		RT	20	10-fold cross-validation		Liu et al. (2021)
		RT	15	Leave-one-subject-out cross-validation		Ko et al. (2020)
		Self-reported fatigue questionnaire	35	leave-one-out (LOO) cross-validation		Min et al. (2021)
	Driving Simulation & Real Driving	KSS	20	Cross-validation		Du et al. (2021)
	Driving Simulation & PVT	SSSQ	40	Leave one out cross-validation		Dimitrakopoulos et al. (2018)
	Real driving	SSS	10	Ten-fold cross-validation		Zhang et al. (2021a)
	Visual Stimuli	subjective fatigue scale	11	10-fold cross validation	General user	Peng et al. (2021)
	Psychomotor vigilance test (PVT)	SSSQ	40	10-fold cross-validation		Qi et al. (2020)
	Maximal Voluntary Contraction(MVC)	SSS, KSS, SPC, SFS, and Borg Rating Scale (CR-10)	20	5 double cross-validations	Athletee	Yang and Ren (2019)
	Flight Simulation	NASA-TLX scale and KSS	40	10-fold cross-validation	Pilot	Wu et al. (2022)
EOG	The N-back task	VAS	30	10-fold cross-validation	Worker	Kolodziej et al. (2020)
ECGMulti	Treadmill exercise	RPE scale	80	10-fold cross validation	Athlete	Ni et al. (2022)
	Flight Simulation	SPF 7-Level scale	30	K-fold cross validation ($K = 10$)	Pilot	Pan et al. (2021)
EMG	Driving Simulation	FAS	15	k-fold cross validation	Driver	Naim et al. (2022)
Gait/ Motion	walk	The 30-item POMS-SF	126	10-fold cross-validation	Athlete	Kadry et al. (2022)
	Walking and Exercise	VAS	8	Leave-One-Subject-Out Cross-Validation		Aoki et al. (2021)
	Exercise	1 to 10 scale	14	Nested K-fold cross-validation		Jiang et al. (2021)
	Manual material handling (MMH)	RPE 6–20 scale	14	5-fold cross-validation	Worker	Baghdadi et al. (2018)
Multi	Driving Simulation	Feedback 1–5 scale	26	five fold cross validation	Driver	Choi et al. (2018)
		eye movement information	23	fivefold cross-validation		Zhang et al. (2022)
		RT	22	cross-validation		Wang et al. (2019)
		KSS	16	cross-validation (CV): intra-subject and inter-subject CV.		Min et al. (2022)
	Auditory stimuli	FAS	17	Ten-fold cross-validation (CV)	Worker	Ramírez-Moreno et al. (2021)
	Ship navigation task	KSS	11	Nested 20-fold cross validation	Ship Captain	Monteiro et al. (2020)
	daily life activity	FAS and VAS	28	cross-validation	General user	Luo et al. (2020)
	Pilates movement training	RPE scale	20	Monte Carlo cross-validation (MCCV)	patient	Li and Chen (2022)

*Karolisnka Sleepiness Scale (KSS), NASA-Task Load Index (NASA-TLX), Stanford Sleepiness Scale (SSS), Reaction Time (RT), Short Stress State Questionnaire (SSSQ), Samn-Perelli Scale (SPC), Subjective Fatigue Rating Scale (SFS), Profile of Mood States-Short Form (POMS-SF), visual analogue scale (VAS), Fatigue Assessment Scale (FAS), Rating of Perceived Exertion(RPE)

(PVT) (Qi et al., 2020; Dimitrakopoulos et al., 2018), and N-back (Kolodziej et al., 2020) protocols to induce mental fatigue of the participants. The duration of these experimental protocols are quite shorter than other experimental protocols such as driving and flight simulation. Ramírez-Moreno et al. (2021) conducted the experiment for only five minutes, which is also the shortest experimental protocol compared to other studies.

Simulation protocol is found to be the most popular form of data collection protocol. The first reason is that it is very easy to setup and can be utilized in collecting several physiological signals simultaneously. Secondly, several studies have argued that the experimental protocol used in real-world scenarios poses safety risks to participants. Lastly, unlike real environment, the signals collected in simulated environments are less susceptible to noise, which makes the signals less corrupted and easier to analyze.

Table 6
Summary of signal characteristics (sampling rate and number of channels) used by existing EEG based mental fatigue detection studies.

Sampling Rate (Hz)	Number of channels	Reference
128	14	Wang et al. (2021, 2021a)
	19	Yang and Ren (2019)
160	64	Wu et al. (2022)
200	61	Zeng et al. (2021, 2020)
250	3	Liu et al. (2021)
	24	Xu et al. (2021, 2021, 2021)
500	32	Zhang et al. (2021b)
512	1	Ko et al. (2020)
	64	Dimitrakopoulos et al. (2018, 2020)
600	16	Peng et al. (2021)
1000	2	Luo et al. (2019) Min et al. (2021)
	12	Sheykhivand et al. (2022, 2022)
	30	Dang et al. (2021); Gao et al. (2019); Yang et al. (2021); Hu and Min (2018); Gao et al. (2019); Wang et al. (2018)
	40	Cai et al. (2019)
~	16	Du et al. (2021)

4.2. Controlled environment

There are 7 studies out of 43 studies conducted in controlled environment. In majority of the exercise based controlled environment studies, the duration of the experiment is controlled based on participants feedback of being exhausted or not. Among them (Yang & Ren, 2019; Ni et al., 2022; Aoki et al., 2021; Jiang et al., 2021), the data collection protocol followed a set of exercises (walking, running, squatting, high-knee jack and corkscrew toe touch, maximum voluntary contraction) until the participant reported the state of exhaustion. Only Kadry et al. (2022) and Li and Chen (2022) conducted the experiment for a fixed duration of 2 minutes and 15 minutes, respectively. Apart from exercise, Baghdadi et al. (2018) conducted a manual material handling protocol to mimic manufacturing tasks for three hours to collect gait data.

Studies that mainly aim to collect data for physical fatigue investigation are more targeting controlled experimental protocol. Similar to the studies used simulation protocol, studies in this group also preferred to avoid real-world data collection challenges. A lab based experimental protocol provides a better control in ensuring the safety of the participants as well as minimizing the impact of external factors in fatigue induction.

4.3. Real environment

Unlike simulated environment and controlled environment data collection protocols, very limited studies were observed to have conducted a real world experimental protocol for data collection that is influenced by the real world situations. Zhang et al. (2021a) and Du et al. (2021) are the only studies that collected real driving data. To ensure the safety of the participants, Du et al. (2021) conducted the outdoor driving under close monitoring, and if a participant was scored 8 or more in KSS assessment they were not allowed to drive outdoor. The study of Luo et al. (2020) was interesting where data were collected from 28 participants in daily activity without setting any protocol. The participants were free to continue their daily activity keeping the wearable device attached to the body. However, due to lack of control, the authors reported missing samples due to the loss of skin contact and subjects not wearing the device.

Due to the lack of control over the signal, it's still a challenge to get clean data in real-world experimental scenario. Aspects like missing sample and noise in signal require additional prepossessing of the signal. Although several techniques are available to impute missing sample, most studies preferred to avoid such additional steps.

5. Fatigue detection model

Objective fatigue detection is the process where machine learning

Table 7
Summary of Research environment in the past fatigue detection studies (Simulated environment/ controlled environment/ real environment), and their experimental duration.

Research environment	Experimental task/stimuli	Experimental duration				
		<40 min	40-120mins	120–360mins	until fatigued	others
Simulated Environment	Driving	Wang et al. (2021)	Cai et al. (2019); Dang et al. (2021); Dimitrakopoulos et al. (2018); Z. Gao et al. (2019); Z.K. Gao et al. (2019); Hu & Min, (2018); Ko et al. (2020); Min et al. (2022); Naim et al. (2022); Sheykhivand et al. (2022); Sheykhivand et al. (2022); H. Wang et al. (2019, 2021); H.T. Wang et al. (2021); P. Wang et al. (2018); Xu et al. (2021); Yang et al. (2021); X. Zhang et al. (2022); X.W. Zhang et al. (2021b)	Choi et al. (2018); Du et al. (2021); Liu et al. (2021); Min et al. (2021); Zeng et al. (2020, 2021)	Luo et al. (2019)	
	Flight			Pan et al. (2021); Wu et al. (2022)		
	Ship		Monteiro et al. (2020)			
	Auditory/Visual stimuli	Peng et al. (2021); Ramírez-Moreno et al. (2021)				
	N-back		Kolodziej et al. (2020)			
	PVT	Qi et al. (2020)	Dimitrakopoulos et al. (2018)			
Controlled Environment	Exercise	Kadry et al. (2022)	Li and Chen, (2022)		Aoki et al. (2021); Jiang et al. (2021); Ni et al. (2022); Yang & Ren, (2019)	
	Object Handling			Baghdadi et al. (2018)		
Real Environment	Driving			Du et al. (2021); Zhang et al. (2021a)		
	Daily Life					Luo et al. (2020)

model is used to measure fatigue levels based on features extracted from physiological signals. The objective fatigue detection systems can be divided into two primary categories based on the type of input. The conventional approach involves the use of features that are manually extracted, while the deep learning-based system is capable of identifying patterns from raw signals to accurately predict an individual's state of fatigue.

Past studies have utilized diverse types of machine learning models such as SVM, Neural Network, Ensemble, Regression model, and statistical model to classify the fatigue state. The neural network based fatigue detection model is found to be most used by 15 studies to classify the fatigue and non-fatigue state followed by SVM used by 11 studies and ensemble model by 9 studies (**Table 8**).

Several types of neural network such as Artificial Neural Networks (ANNs) (Peng et al., 2021; Zeng et al., 2021), Convolutional Neural Networks (CNNs) (Wang et al., 2021, 2021, 2019, 2019, 2021, 2020, 2020) and Recurrent Neural Networks (RNNs) (Aoki et al., 2021; Zhang et al., 2021b; Yang et al., 2021), Generative Domain Adaptation Neural Network (Generative DANN) (Zeng et al., 2021) were found to be used by the studies in finding fatigue state. CNN architecture was found prominent to propose various models such as CNN-LSTM (Sheykhivand et al., 2022; Sheykhivand et al., 2022; Wu et al., 2022); CNN-Attentionp (Xu et al., 2021), Attention-based multiscale CNN-dynamical graph convolutional network (AMCNNDGCN) (Wang et al., 2021), EEG-based spatial–temporal CNN (ESTCNN) (Gao et al., 2019), Recurrence network CNN (Gao et al., 2019). CNN based neural network is leveraged by all these studies to extract features from the raw EEG data, which reduces complexity and increases efficiency. All of these studies recorded to achieve a significant accuracy in fatigue detection.

Apart from mental fatigue detection, neural network based models are also utilized in physical fatigue detection. A multi task learning RecurrentNeural Network (RNN) model was used by Aoki et al. (2021)

Table 8
Fatigue detection accuracy reported in the selected studies based on type of machine learning models and signals used.

Group	Signal	Model	Accuracy	Reference
Neural network	EEG	CNN-LSTM	99.23%	Sheykhivand et al. (2022)
	EEG	DC-LSTM	96.36%	Sheykhivand et al. (2022)
	EEG	CNN-LSTM	91.54%	Wu et al. (2022)
	EEG	RNN	89.28%	Zhang et al. (2021b)
	EEG	Generative DANN	91.63%	Zeng et al. (2021)
	EEG	CNBLS	99.50%	Yang et al. (2021)
	EEG	unified CNNAttention	97.80%	Xu et al. (2021)
	EEG	ANN	87.70%	Peng et al. (2021)
	EEG	AMCNNDGCN	95.65%	Wang et al. (2021)
	EEG	TCRFN	88.00%	Du et al. (2021)
	EEG	CNN	97.37%	Gao et al. (2019)
	EEG	RN-CNN	92.25%	Gao et al. (2019)
	Gait/Motion	RNN	91.50%	Aoki et al. (2021)
	Gait/Motion	CNN	94.00%	Jiang et al. (2021)
	ECG, EMG, ST, EEG, and eye tracker.	CNN	82.00%	Monteiro et al. (2020)
SVM	EEG,EOG	SVM(linear)	99.48%	Min et al. (2022)
	EEG	cubic SVM	97.20%	Zhang et al. (2021a)
	EEG	SVM	90.00%	Wang et al. (2021)
	EEG	SVM	96.76%	Wang et al. (2021)
	EEG	SVM	90.00%	Yang and Ren (2019)
	EEG	SVM	95.37%	Luo et al. (2019)
	EEG	SVM	99.09%	Cai et al. (2019)
	EEG	SVM	97.40%	Dimitrakopoulos et al. (2018)
	Gait/Motion	SVM(RBF kernel)	90.00%	Baghdadi et al. (2018)
	ECG, sEMG	IPOS-SVM	87.83%	Li and Chen (2022)
	EEG,EOG	RVM	99.10%	Wang et al. (2019)
Ensemble	EEG	Hybrid (LR + ELM + LightGBM)	94.20%	Min et al. (2021)
	EEG	RF, BP_Adaboost	92.70%	Liu et al. (2021)
	EEG	EN	95.28%	Dang et al. (2021)
	EEG	EN	94.00%	Wang et al. (2018)
	EEG	GBDT	94.00%	Hu and Min (2018)
	ECG	LightGBM	85.50%	Ni et al. (2022)
	Gait/Motion	GB	74.30%	Kadry et al. (2022)
	PPG,TEMGSR,ACC,gyro	SVM-MWV, SVM-WTA	98.43%	Choi et al. (2018)
	Accelerometer, PPG, ST, GST, Barometer	RF	71.85%	Luo et al. (2020)
Statistical	EEG	MLR	68.00%	Ko et al. (2020)
	EEG	RVR	RT MAE = 0.0255 and TOT slop MAE = 0.0063	Qi et al. (2020)
	EMG	LDA	95%	Naim et al. (2022)
	EOG	QDA	93.00%	Kolodziej et al. (2020)
	EEG, HR, HRV and EDA	MLR	88.00%	Ramírez-Moreno et al. (2021)
Others	EEG	InstanceEasyTL	88.33%	Zeng et al. (2020)
	ECG	LVQ	81.94%	Pan et al. (2021)
	EEG,EOG	GMC-KSRR, GMC-MKSRR	91.56%	Zhang et al. (2022)

*Convolutional neural network Long short-term memory (CNN-LSTM), Deep convolutional long short term memory (DC-LSTM), Artificial neural network (ANN), TSK-type convolution recurrent fuzzy network (TCRFN),convolutional neural networks (CNNs), Recurrence network CNN (RN-CNN), Recurrent neural network (RNN), Convolutional neural network (CNN), Support vector machine (SVM), Improved particle swarm optimization-support vector machine (IPSO-SVM), Complex network based broad learning system (CNBLS, Logistic regression (LR), Extreme learning machine (ELM), Light gradient boosting machine (LightGBM), Gradient boosting(GB), Random forest (RF),Gradient boosting Decision Tree(GBDT), Expanded SVM with Winner-takes-all (WTA) and max-wins voting (MWV), Linear vector quantization (LVQ), Generalized minimax-concave (GMC) penalty-based SR with kernel method (GMC-KSRR, GMC-MKSSR), Ensemble (EN), Attention-based multiscale CNN-dynamical graph convolutional network (AMCNNDGCN) to analyze the gait cycle data. Jiang et al. (2021) used CNN to investigate the physical fatigue level from three exercises (squat, jack, and corkscrew) using IMU data. In contrast, Luo et al. (2020) utilized a multi-modal fatigue detection using CNN to detect both mental and physical fatigue of the participants.

The research in fatigue detection has seen a significant focus on using neural network based solutions in contrast to traditional two-stage fatigue detection approaches. Unlike neural network based fatigue systems, the two stage fatigue detection system requires extracting feature manually from the raw data to feed the model for training and testing. One of the key reasons is the learning capability of features without any intervention, thus reducing the time complexity and bias of manual features identification and selection. A network like CNN due to its ability to automatically learn and extract time-series data feature is found to be the most popular. However, they also have some drawbacks due to computational costs based on the use of network layers. Network based architectures rely highly on the tuning of hyper-parameter where finding optimal parameters can be challenging.

SVM, which is a type of supervised machine learning model, is best suitable for regression and classification tasks. Several variant of SMV such as linear kernel SVM (Wang et al., 2021), cubic SVM (Zhang et al., 2021a), SVM with radial basis function (RBF) kernel (Dimitrakopoulos et al., 2018; Baghdadi et al., 2018), and Relevent vector machine (Wang et al., 2019) are found to be used in several studies. Although most of the studies used binary classification technique using SVM, a study using multi-class SVM classification was also found (Li & Chen, 2022). Studies that utilized a more traditional approach in extraction of features from EEG data found to be using SVM as classification task to detect mental fatigue and non-fatigue state (Zhang et al., 2021a; Wang et al., 2021; Wang et al., 2021; Yang & Ren, 2019; Luo et al., 2019; Cai et al., 2019; Dimitrakopoulos et al., 2018). The study by Baghdadi et al. (2018) used rbf kernel SVM to classify physical fatigue. To classify fusion features from multi-modal signals, different varient of SVM were utilized. Min et al. (2022) and Wang et al. (2019) used linear SVM and RVM to detect driver's metal fatigue, respectively. On the other hand, Li and Chen (2022) trained SVM with particle swarm optimization alogrithm to achieve better accuracy in multi-class physical fatigue classification of the athletes. SVM is the most popular classifier for solving the regression problem due to its high generalization capability, which fits irrespective of the signal modality. Current research is shifting away from the use of SVM-like classifiers because of the complexity of manual feature extraction and selection process. However, in most of the studies, SVM is used as model validation, which acknowledges the prominence of SVM in any modality of fatigue detection.

Ensemble classifiers are a combination of multiple-individual classifiers that have been in active use in fatigue detection studies. Studies found using gradient boosting(GB), LightGBM, bagging, BP_Adaboost multi-class SVM, subspace discriminant, and random forest(RF) for fatigue classification. Ni et al. (2022) and Kadry et al. (2022) used LightGBM and Gradient boosting classifier, respectively, for physical fatigue detection. Choi et al. (2018) used a multi-class SVM classifier for driver's multiple mental conditions detection. To classify EEG based mental fatigue, Min et al. (2021) used a hybrid classifier combining logistic regression, Extreme learning machine, and lightGMB. Liu et al. (2021) achieved better accuracy using BP_Adaboost in independent-subject classification, but failed to achieve a better result in using RF classifier in inter-subject fatigue classification. Similarly both Wang et al. (2018) and Hu and Min (2018) used GB classifier and Dang et al. (2021) used subspace discriminant ensemble classifier to achieve a better accuracy in fatigue classification. Ensemble classifier combines the complementary information from multiple base classifiers which reduces the chance of over-fitting and will ultimately result in achieving high accuracy.

Few studies were observed to utilize either a single machine learning based fatigue detection model or hybrid machine learning model. Pan et al. (2021) used a supervised machine learning model named Linear vector quantization (LVQ) to classify pilot's mental fatigue state from ECG signal. Zhang et al. (2022) used a non-linear extension of the Sparse Representation Regression (SRR) model to identify fatigue using EEG and EOG signal fusion. A transfer learning based machine learning model by Zeng et al. (2020) improved its predecessor by achieving better accuracy in cross-subject mental fatigue detection. Lastly, a hybrid model that combines CNN and RNNs with fuzzy logic was proposed by Du et al. (2021) to detect fatigue from EEG signals.

Apart from different machine learning based models, fatigue detection studies also found utilizing statistical models like Multiple Linear Regression (MLR), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis, and Relevance Vector Regression (RVR). Naim et al. (2022) used LDA, which is a type of discriminant analysis and linear classifier on EMG data to classify driver muscle fatigue. On the other hand, Kolodziej et al. (2020) used another form of discriminant analysis but non-linear classifier named QDA to identify the mental fatigue state of office worker using only EOG signal. Qi et al. (2020) applied RVR, which is a linear regression method to detect mental fatigue using EEG functional connectivity. MLR is also found to be used in classifying multi-modal signals by Ko et al. (2020) (EEG-EOG fusion features) and Ramírez-Moreno et al. (2021) (EEG, HR, HRV, EDA and ST features).

In terms of the performance of the different classifiers, it appears that neural networks, SVM, and ensemble classifiers generally achieve higher accuracy compared to other methods, with some articles reporting an accuracy as high as 99.5%. However, it is difficult to make generalizations about the performance of the different classifiers as the performance can vary depending on the specific implementation and the data used. SVM based studies achieved an average of 96.76% accuracy in comparison to 92.25% and 94% by neural network and ensemble based studies (Table 8). Although neural network provided automatic feature learning and classification, SVM due to its high robustness still found to be the best classifier in fatigue detection research. To understand whether choosing certain model has impact on accuracy over others, we used comparison approaches. Fig. 5 illustrates the min, max, and median accuracy for each fatigue classification group. Neural network and SVM showed better accuracy over other groups. One of the existing hypothesis of using multi-modal approach is the rich presence of fatigue information in contrast to uni-modal approach. However, as depicted in Fig. 6a, only SVM and Ensemble models achieved better accuracy in comparison to uni-modal approaches. Interestingly, the multi-modal approach that included EEG signal generally achieved better accuracy than others. Since majority of the studies used EEG signal, a comparison of accuracy of different models for EEG versus non-EEG signals was also conducted (Fig. 6b). The conventional wisdom holds that EEG signal has the rich presence of fatigue information, which enables machine learning models to find patterns more efficiently and subsequently achieve better accuracy. However, EEG based approaches are also considered very complex in design and implementation in-contrast to other physiological parameters. Approaches with alternative signals for EEG can be more practical and less complex to design. Due to the lack of sufficient number of past studies in each of the compared categories, it was difficult to make a concluding remark on which model is best for which modality or physiological parameter.

Although most of the studies aimed in achieving high accuracy in predicting the fatigue and non-fatigue state using the above discussed machine learning models, most of them only used subject dependent data for training and testing their models. Machine learning model can learn patterns with relatively high accuracy when the training and testing data are subject dependent due to the data leakage mechanism. Due to this phenomena, most of these models are able to achieve such high accuracy on already known dataset, but are unable to achieve such accuracy when new subject independent data is provided. Out of 43 studies, only 5 studies used both subject dependent and subject independent training and testing of the model. Using SVM on multi-modal data, Min et al. (2022) and Choi et al. (2018) achieved 98.43% and

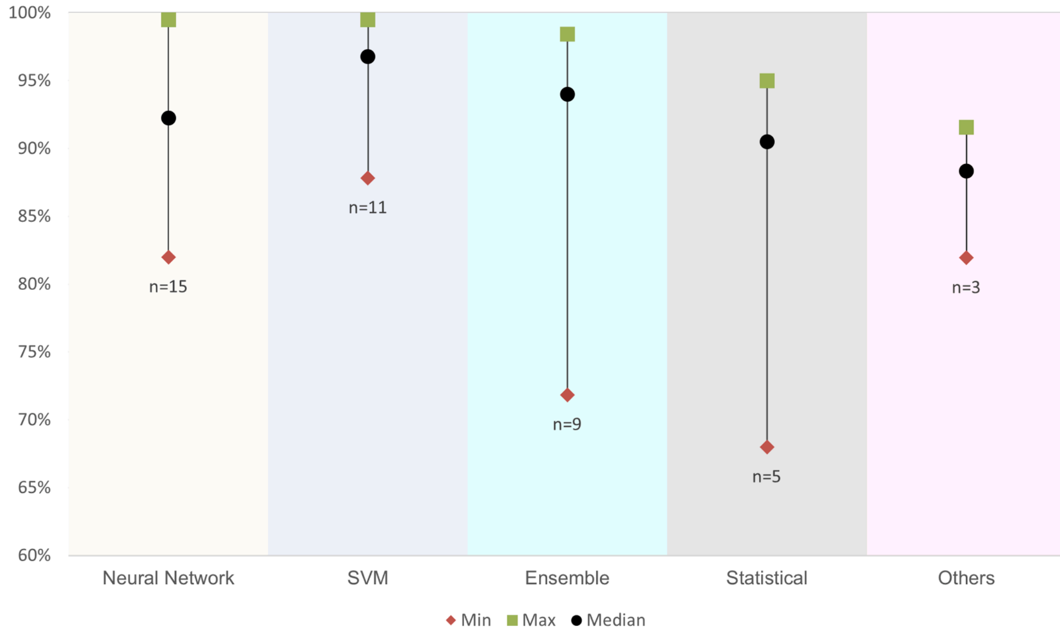

Fig. 5. Accuracy (*Minimum*(*Min*) −*Median* −*Maximum*(*Max*)) of machine leaning based fatigue detection models extracted from the selected studies and grouped by the used model (Neural network, SVM, Ensemble, Statistical, and Others). *n is the number of studies found in each group.

99.48% in subject dependent approach, but achieved only 68.31% and 71.98% accuracy, respectively, in subject independent classification. In mental fatigue classification using EEG and EOG, Zeng et al. (2021) and Kolodziej et al. (2020) achieved 92.70% and 93% in subject dependent training whereas achieved 77.13% and 89% in subject independent approach, respectively. Similar drop in accuracy also reported by Aoki et al. (2021) where subject-independent approach achieved 86% in-contrast to subject-dependent, which achieved 91.5% accuracy. The finding by these studies indicated the drop in accuracy by the model in subject independent approach. Therefore, the reported high accuracies in subject dependent models that constitute most of the past studies are not reliable.

6. Future directions

The following section provides insights into the reviewed studies highlighting the limitations and research gaps and future directions in the field of fatigue detection using physiological signal and machine learning techniques.

6.1. Signal modality

The recent research has still focused on the use of the uni-modal approach in detecting fatigue across different domains. Based on our review, 35 out of 43 selected studies used a uni-modal approach to detect physical and mental fatigue. The majority of these uni-modal studies were related to mental fatigue detection.

To identify mental fatigue, the use of EEG was proven efficient by the previous studies in lab-based tests and was used by the majority of them. However, due to the invasive nature of the data acquisition process, EEG is not considered easy to integrate in providing fatigue monitoring solutions in real-life conditions (Borghini et al., 2014). Although few studies investigated the use of fewer channels or certain regions such as frontal, all of these studies were conducted in lab-based data collection protocol that put significant doubt on the actual performance of the proposed models in the real world where the device should be worn for a longer time. Thus, further research should warrant:

- reducing number of EEG channels with an aim to improve comfort of the device.
- determining the generalization capacity of the reduced channel-based solutions by further validating in real-world settings.
- identifying alternative physiological signals that will provide more practical and feasible solutions for mental fatigue detection.

Besides uni-modal approaches, there are studies focusing on the use of multi-modal signals. The benefit of multi-modal solution is that it can add complementary information especially in the absence of multi-channel EEG signal (Laurent et al., 2013). However, little attention has been paid on systematic investigation of potential alternative physiological signals that can be used for different types of fatigue detection. Therefore, future studies should focus on:

- determining the set of physiological signals that can be used in combination with the reduced channel EEG solution to keep the accuracy at an acceptable level with reduced false alarms.
- determining a set of physiological signals that can be used without EEG and without compromising the performance in mental fatigue detection.
- validating the proposed solutions in real-world scenarios rather than only in lab or controlled environment settings.

Although Multi-modal approach has clear advantages over uni-modal approach, it is faced with an inherent challenge to combine multiple sensors in a single device that is capable of collecting various physiological signals without causing discomfort for the users. Even with the advancement of technology and the innovation in modern lightweight devices, there are not many studies focusing on investigating real-world fatigue detection scenarios.

6.2. Experimental environment

From the review, it was found that apart from three studies, the rest of them used a lab-based data collection system. Many studies argued to have taken a lab-based experimental setup to avoid the risks and uncertainties associated with setting a real-world or outdoor setup. Real-world physiological data collection is prone to artifact, particularly in

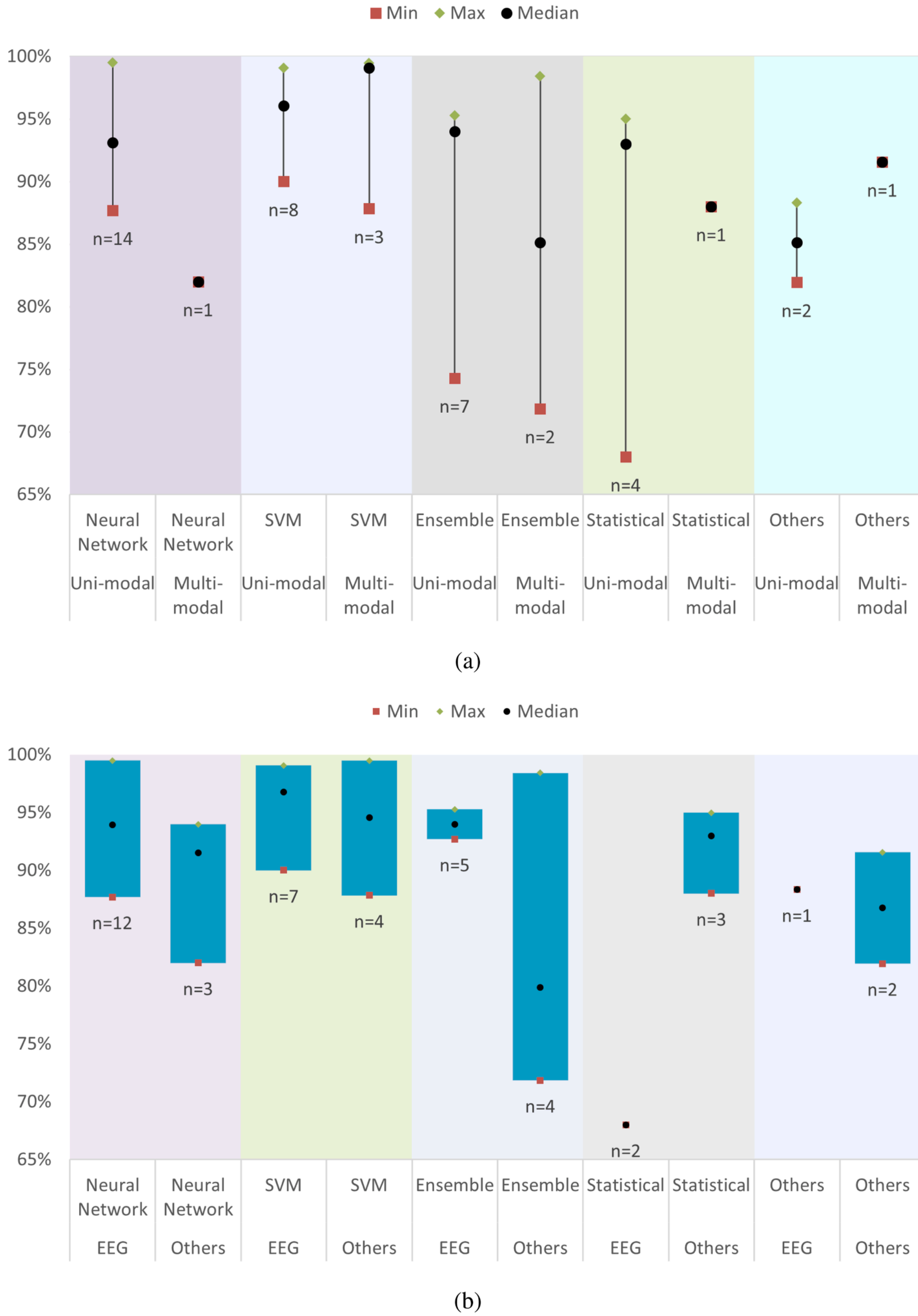

Fig. 6. Accuracy (*Minimum*(*Min*) –*Median* –*Maximum*(*Max*)) of machine leaning based fatigue detection models extracted from the selected studies and grouped by the used model (Neural network, SVM, Ensemble, Statistical, and Others): (a) by signal modality (b) by model in comparison between EEG and other physiological parameters.*n is the number of studies found in each group.

mobile monitoring where motion can significantly corrupt the data (Jebelli et al., 2018; Anwer et al., 2021). The lab-based data collection can produce less noise, which makes signal processing easier. However, fatigue detection research must be expanded outside of the lab to collect data from real-life situations and provide practical solutions to control fatigue in daily jobs. This will not only challenge the researchers in processing noisy data, but also force them to remodel their approach to find an acceptable result. Future research can also focus on:

- developing noise removal techniques for the data collected in real-world scenarios.
- investigating the potential use of inertial sensors (accelerometer, gyroscope, etc.) for filtering noise from the collected physiological signals to improve the performance of the overall solution.
- remodelling of offline fatigue detection models using real-world data and advanced machine learning approaches including deep learning and reinforcement learning.

6.3. Bias in physiological signal selection

Based on our review results, another issue is the presence of bias in the selection of physiological signals for fatigue detection. The past research tended to take a safe path by choosing the most used signals for fatigue detection. If the target is mental fatigue detection, EEG has been mainly used by past studies as a uni-modal approach or in combination with other physiological signals as a multi-modal solution. A similar trend is observed in physical fatigue detection where EMG, ECG or gait has been used in a uni-modal approach in most of the past studies. This indicates a bias in the selection of physiological signals without investigating the suitability of other alternative signals. The future research should focus on:

- investigating individual physiological signals and their correlation with different fatigue types to remove bias in physiological signal selection.
- determining the set of physiological signals that are required for systematically achieving acceptable performance.

6.4. Fatigue modeling

Past studies have used different machine learning models including SVM, neural network, ensemble classifier, statistical models, and so forth to detect fatigue. Current studies suffer from several model design issues such as (1) data leakage, (2) computational cost, (3) off-line processing, and (4) using only physiological parameters for determining fatigue, where many other environmental factors, such as heat, noise, latitude, and so forth can be a significant contributor for inducing fatigue in real-world scenarios. Future research should focus on:

- designing machine learning models that prohibit data leakage such as using part of test data in training. Otherwise, the model may show very high accuracy but fail to show acceptable performance after deployment.
- designing a continuous fatigue monitoring model rather than an offline one, since the second one cannot be used for intervention design in real-world scenarios.
- designing a more comprehensive fatigue detection model by adding factors other than instantaneous physiological responses.

7. Challenges of objective fatigue detection in real work environment settings

Several challenges are to be considered in order to provide practical fatigue monitoring solutions using physiological signals. In this section, we will try to highlight some of the challenges that need to be addressed in future research to pave the ways for practical fatigue monitoring systems.

7.0.1. Channel reduction of EEG

For mental fatigue detection, the most common and reliable source of data is collected via EEG devices. However, due to the intrusive nature of the EEG device, its is yet not practical to be used in outdoor environment for an extended period of time. Research on reducing the number of channels and finding the optimal amount of data is much needed to come up with a lighter device that can effectively provide the required data for mental fatigue detection.

7.0.2. Multi-modal device

Based on surveys, people are not comfortable wearing more than one device at a time. Having a single device that can collect all the required physiological parameters is a step forward towards the practical implementation of fatigue detection systems in the real world. Although several current devices are able to collect multi-modal signals such as PPG, EDA and accelerometer, having devices capable to collect other important combinations of physiological parameters (e.g., EEG + HR) is an important need that needs to be addressed.

7.0.3. Synchronizing multiple sensors

Unlike multi-modal devices where data collection from multiple sensors can be done simultaneously, multi-device where data are collected from multiple devices can be challenging. To develop a near real-time fatigue monitoring system, data synchronization is vital to train the model to predict the fatigue state. The data collection pipeline need further attention in research to find an easy to integrate solution, which can handle multiple sensor data collection in sync.

7.0.4. Modeling multiple dimensions of fatigue

Most of the past studies have focused on one of the fatigue contributors (i.e., physical or mental exhaustion). However, there are several other contributors (e.g., heat, noise, light, sleep, etc.) that should be taken into account to develop an accurate fatigue detection model. The manifestation of fatigue is highly driven by external factors that at the current level of research are not in consideration by the research community. However, the major aspect of a practical fatigue monitoring system is to have a comprehensive model that not only takes the physiological changes into consideration, but also can predict the effect of external factors that can affect the level of fatigue.

7.0.5. Interactions between physical and mental fatigue

Almost all past studies have studied either physical or mental fatigue. However, in the real world, people are affected by both types of fatigue simultaneously. It is important to know how physical and mental fatigue interact. Without knowing those interactions, we cannot detect the level of fatigue and design efficient mitigation strategies. Therefore, further research into the interaction between physical and mental fatigue is required.

8. Conclusion

Fatigue due to its direct association with a broad array of physical and psychological effects as well as life-threatening impacts such as accidents and injuries need significant attention in terms of automatic monitoring and practical solutions. Fatigue detection and monitoring using physiological signals have been proven to be reliable and objective approaches. In this study, a review of the most recent studies in the field of fatigue detection using physiological signals was conducted to identify the state of art solutions and existing research gaps. For the selection of suitable studies, a set of selection criteria was used to avoid bias in existing studies, which ensured a common standard in quality among the selected studies. The final review was conducted on 43 peer-reviewed journal articles focusing on three main topics: (1) signal modality; (2) experimental environment; and (3) fatigue detection model. A comprehensive discussion and future directions based on these topics were also presented in Section 6. Furthermore, five challenges of the fatigue detection in the real-world scenarios have been identified including: (1) channel reduction of EEG; (2) multi-modal device; (3) synchronizing multiple sensors; (4) modeling multiple dimensions of fatigue; and (5) interaction between physical and mental fatigue. These challenges will provide a scope for the researchers and practitioners to further research towards the practical fatigue detection solution. This review emphasizes the need for further investigating practical fatigue detection approaches to control the risk of fatigue in real-world workplaces.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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